

SAWDA: A Benchmark for Intent-Preserving Translation in Cross-Border Trade Communication

Murtaza Nikzad*

[TBD: co-authors: R. Ramanujan? J. Leigh (see acknowledgments note)]

Working draft — August 2026

Abstract

Machine translation (MT) systems and large language models (LLMs) now produce lexically accurate translations for most high-resource language pairs, yet everyday cross-border communication continues to fail in a way that accuracy metrics do not capture: the words are translated and the intent is lost. We study this failure in the setting where its cost is most concrete, trade communication between Persian-speaking (Farsi/Dari), English-speaking, and Chinese-speaking business partners. We introduce SAWDA, a benchmark of 60 culturally loaded, native-validated trade-communication items in Dari/Farsi and English, built from author-written seeds, the pragmatics literature, and validated synthetic augmentation. Each item pairs a source utterance and its relationship context with a literal rendering, an intent-preserving rendering, and an annotation of the pragmatic content at risk. In a pilot study, we compare six open-weights configurations across three model families: a dedicated low-resource MT model (NLLB-600M), a frontier-class open LLM (Kimi K3) under naive, audience-targeted, and instructional prompting, and a 20B self-hostable model (gpt-oss-20b) under naive and audience-targeted prompting. In 180 blind judgments by a native Dari speaker, no system reliably preserved speaker intent: the best configuration fully preserved intent in 9 of 30 judgments. Explicit cultural instruction raised preservation over audience-targeting alone (9/30 vs 5/30 preserved) but also grew the catastrophic tail (5/30 vs 2/30 inverted or deleted), a both-tails pattern that refines, without statistically confirming at pilot scale, the threshold effect reported for appropriateness by [Tenzer et al. \[2026\]](#). Our pre-specified direction hypothesis reversed: failures concentrated in English-to-Dari generation (15/90 vs 5/90 judgments inverted or deleted, exact $p = .031$), driven by broken or wrong-language output, suggesting that generating Afghan Dari, not comprehending it, is the current bottleneck. The 20B model with a one-line audience wrapper was not separable from the frontier model on human ratings at this sample size, at 2–3% of its API cost. Finally, two LLM-judge families topped out near 49% exact agreement with native-speaker labels (chance-corrected $\kappa \leq 0.17$) against a pre-specified 80% bar, so all evaluation numbers are human-labeled: for this task, the native-speaker judge is not yet automatable, which is itself the strongest argument for the benchmark.

1 Introduction

Consider a real exchange between two business partners, one writing in Chinese, one reading in English. The Chinese message, rendered by the messaging platform’s built-in translator, arrived as an accusation: “then you didn’t provide assistance.” The same message, translated with attention to its context, was a request: “do what you did before.” Nothing in the sentence was mistranslated

*Davidson College. Contact: munikzad@davidson.edu

at the level of words. What failed was the transfer of intent, and the failure nearly damaged the relationship the sentence was trying to maintain.

Failures of this kind are systematic rather than accidental. High-context communication cultures [Hall, 1976] encode obligation, refusal, deference, and face-work in forms that literal translation strips away. Persian *taarof* (also transliterated *ta’arof*) ritualizes offer and refusal so thoroughly that a literal “no” frequently means “ask again” [Beeman, 1986, Koutlaki, 2002]. Chinese business communication routes disagreement and obligation through indirection and face management [Hwang, 1987, Hong and Tu, 2026]. When such utterances cross a language border through a literal translator, the pragmatic layer is silently deleted, and neither party learns that a deletion occurred.

Recent work has begun to measure the cultural competence of LLMs, but the measurement has concentrated on monolingual behavior: whether a model can act appropriately within a culture [Gohari Sadr et al., 2025, Moosavi Monazzah et al., 2025], recall cultural knowledge [Myung et al., 2024, Rao et al., 2024], or reason over culture-specific commonsense [Sun et al., 2024]. The translation setting, where cultural competence must survive a language boundary, has received attention only recently and only for a single pair, with Tenzer et al. [2026] demonstrating for English-to-Japanese workplace email that prompting strategy substantially changes the cultural appropriateness of LLM translations. No benchmark exists for the triangle we study, and to our knowledge none exists for any pair involving Persian varieties, despite Dari’s status as a low-resource language spoken by tens of millions [NLLB Team et al., 2022].

We make four contributions. First, we introduce SAWDA¹, a benchmark of 60 native-validated items for intent-preserving translation between Dari/Farsi and English in trade settings, constructed by a pipeline of author-written seeds, literature mining, LLM augmentation, and native-speaker validation, following the construction methodology of Gohari Sadr et al. [2025]; the Chinese arm of the intended triangle is designed but left to future work. Second, we define an evaluation protocol that scores translations on intent preservation rather than lexical adequacy, using blind native-speaker judgment on a three-point intent scale alongside a seven-point appropriateness scale, adapting the two-level design of Tenzer et al. [2026]. Third, we report a controlled pilot comparison of six open-weights configurations, with three findings: no system reliably preserves intent (best: 30.0%); cultural instruction raises preservation while its catastrophic tail also grows, with the added failures traced to generation breakdown rather than pragmatic overreach; and the failure asymmetry runs opposite to our pre-specified hypothesis, concentrating in Dari generation rather than Dari comprehension. Fourth, we show that two LLM-judge families fail to reproduce native-speaker intent labels (near 49% exact agreement, $\kappa \leq 0.17$, against an 80% bar), suggesting that the distinctions this task turns on are not currently automatable and motivating human-labeled benchmarks in this space.

2 Related Work

2.1 Cultural competence benchmarks for LLMs

The nearest neighbor to our work is TAAROFBENCH [Gohari Sadr et al., 2025], the first benchmark for the Persian politeness system, comprising 450 native-validated role-play scenarios across twelve interaction topics. Its findings motivate ours directly: frontier LLMs performed 40 to 48 points below native speakers when *taarof* was culturally appropriate, and responses rated polite by standard politeness metrics frequently violated *taarof* norms, evidence that Western-derived evaluation frameworks miss culture-specific pragmatics entirely. Notably, supervised fine-tuning and Direct

¹*Sawda* is the Dari word for trade.

Preference Optimization [Rafailov et al., 2023] improved cultural alignment by 21.8 and 42.3 points respectively, establishing that the deficit is addressable with targeted data. SAWDA differs in task: TAAROFBENCH asks whether a model can behave appropriately within Persian interaction; we ask whether intent survives translation out of it.

The Persian evaluation ecosystem has expanded beyond politeness to story-driven cultural understanding [Moosavi Monazzah et al., 2025] and alignment benchmarking [Pourbahman et al., 2025]. For Chinese, CHARM [Sun et al., 2024] measures culture-specific commonsense reasoning, and MILIC-EVAL [Zhang et al., 2025] evaluates pragmatic response selection for China’s minority languages, a task design we adapt. Cross-cultural knowledge benchmarks such as BLENd [Myung et al., 2024] and NormAd [Rao et al., 2024] measure what models know about cultures; our concern is instead what translation destroys. Indirect speech acts, the mechanism underlying many of our items, have been benchmarked monolingually for Korean [Koo et al., 2025], again without a translation dimension.

2.2 Pragmatics and implicature in LLMs

That LLMs struggle with meaning beyond the literal has been documented since Ruis et al. [2022], who showed that models fail zero-shot on conversational implicature of the kind formalized by Grice [1975]. Politeness theory [Brown and Levinson, 1987] predicts where such failures concentrate: in face-threatening acts, where cultures diverge most in how much work the surface form performs. High- versus low-context communication [Hall, 1976] and cross-national value structure [Hofstede, 2001] provide the standard frame for why an utterance adequate in one culture arrives as rude, evasive, or accusatory in another. Persian taarof [Beeman, 1986, Koutlaki, 2002] and Chinese face and favor dynamics [Hwang, 1987] are among the best-described such systems, and business pedagogy for Chinese explicitly trains the interpretation of euphemism and contextual cues that non-native speakers miss [Hong and Tu, 2026]. SAWDA operationalizes these literatures as translation test items.

2.3 Culturally sensitive machine translation

Machine translation evaluation has historically prized adequacy and fluency, and modern massively multilingual systems extend coverage to low-resource languages including Dari [NLLB Team et al., 2022], but coverage is not competence: lexical adequacy can be perfect while pragmatic content is deleted. The study closest to our experimental design is Tenzer et al. [2026], which compared three prompting strategies for English-to-Japanese workplace email translation: a naive translate instruction, audience-targeted prompts specifying the recipient’s cultural background and role, and instructional prompts encoding explicit Japanese communication norms. Using a convergent mixed-methods design that pairs text-level pragmatic coding under the cross-cultural pragmatics scheme of House and Kádár [2021] with evaluations by 85 native-speaker recipients, they find that naive prompts yield only limited adaptation toward target-culture language patterns, audience-targeted prompts produce meaningful improvements, and instructional prompts produce the strongest textual adaptation. Recipient evaluations, however, reveal a threshold effect: instructional prompts do not raise perceived appropriateness or intention to comply significantly beyond audience-targeted prompts, suggesting that lightweight audience cues already capture most of the recipient-perceived gain. That result, in a management venue and for a single high-resource pair, validates the phenomenon, the two-level evaluation method, and the practical promise of context layers over dedicated training. SAWDA extends the question to a partially low-resource pair, replaces constructed polite-register emails with trade communication where the failure mode is categorical rather than

gradational, and adds dedicated MT and self-hostable small models to the comparison, allowing the cost question to be posed directly.

3 The SAWDA Benchmark

3.1 Task definition

Each item is a tuple (u, c, t_ℓ, t_i, a) : a source utterance u in Dari/Farsi or English; relationship context c (roles, prior obligations, the live issue); a literal rendering t_ℓ ; an intent-preserving rendering t_i into the target language; and an annotation a of the pragmatic content at risk. Systems receive (u, c) and produce a translation; evaluation asks which rendering their output realizes.

3.2 Failure taxonomy

Items are tagged with the pragmatic failure they are constructed to elicit, using the identifiers carried in the released data: intent inversion (request read as accusation, as in the motivating example; `intent_inversion`); ritual refusal read literally (taarof; `ritual_refusal_literal`); obligation and face content deleted (`obligation_face_deleted`); register violation (deference stripped or inserted; `register_violation`); indirect refusal read as assent (`indirect_refusal_as_assent`). Our pilot results motivate one addition for scoring translations: generation failure (output empty, syntactically broken, or in the wrong language), which is a distinct failure class from pragmatic inversion and, as Section 5.3 shows, dominates one translation direction.

3.3 Construction pipeline

The pilot release contains 60 validated items in two directions: 40 Dari/Farsi to English and 20 English to Dari/Farsi. Items were sourced in three tranches: 8 seed items written by the first author, a native Dari speaker, from lived trade and family-business communication; 20 items drafted from documented pragmatics phenomena in the taarof and business-communication literatures [Beeman, 1986, Koutlaki, 2002, Hwang, 1987, Hong and Tu, 2026]; and 32 synthetic variants generated from the first two tranches following Gohari Sadr et al. [2025], varying roles, goods, amounts, and relationships while preserving the pragmatic core. All 60 drafted items passed native-speaker validation with inline editing (pass rate 60/60; nothing was discarded, which is itself a limitation of the protocol, since the validator was also the benchmark author). Eighteen items were edited during validation: 9 source utterances, 13 literal renderings, and 11 intent-preserving renderings; pre-edit drafts are released for diffing. All items are fictional and contain no real names, businesses, or identifying details. Six validated items are held out as a few-shot pool for the instructional condition (three are used in its prompt) and are excluded from evaluation for all systems, leaving 54 evaluated items. Figure 1 summarizes the construction and evaluation flow.

3.4 Benchmark statistics

The 60 validated items divide 40/20 by direction (Dari/Farsi→English, hereafter **fa2en**, vs English→Dari/Farsi, **en2fa**) and 8/20/32 by tranche (seed / literature-grounded / synthetic variant). Source utterances average 11.6 words (10.3 for Dari/Farsi sources, 14.1 for English sources), reflecting the short, formulaic register of spoken trade exchanges. Table 1 gives the failure-class composition together with pooled judgment outcomes on the 30-item judged subset, aggregated across the six systems. Indirect refusal read as assent is the hardest class to preserve fully (5.6% of judgments rated preserved), while ritual refusal is the most polarized: systems either land it (36.1% preserved, the highest of any

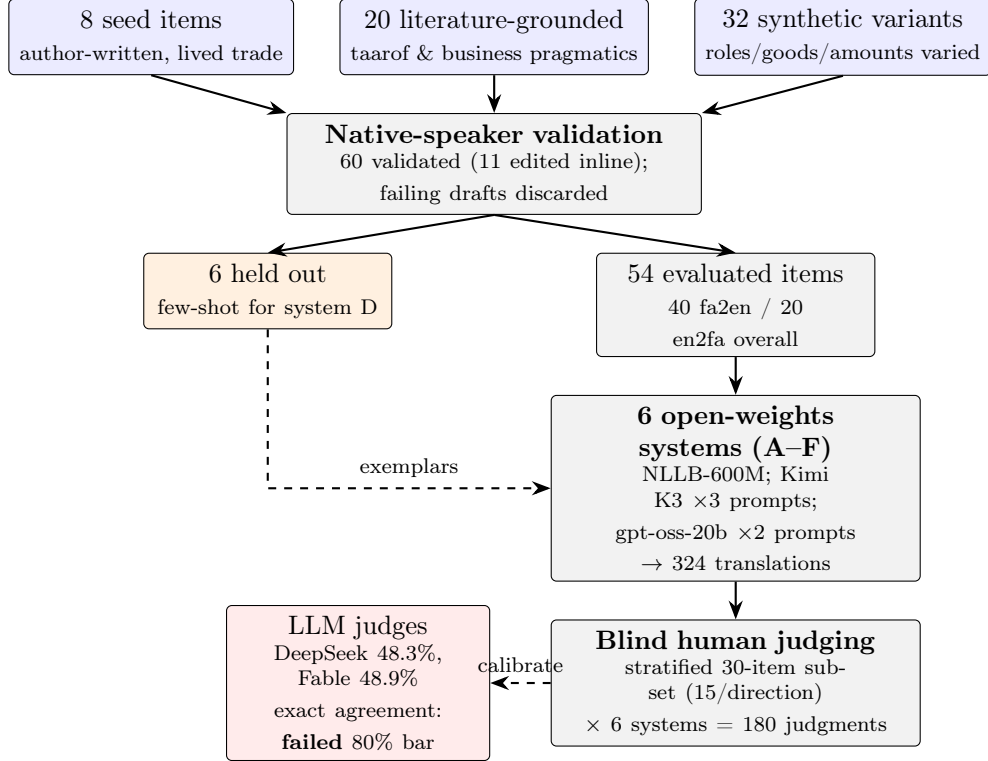


Figure 1: Benchmark construction and evaluation pipeline. All counts trace to the released data files: 60 items in three tranches, 6 held out for few-shot exemplars, 54 evaluated across six systems (324 translations), and a stratified 30-item subset receiving 180 blind native-speaker judgments, against which two LLM-judge families were calibrated and failed the pre-specified 80% exact-agreement bar.

class) or invert it outright (22.2%, also the highest), consistent with taarof being an all-or-nothing translation decision.

4 Experimental Setup

4.1 Systems

Six configurations, all open-weights, all reproducible with a single Fireworks API key (the NLLB baseline ran on a Modal CPU worker): (A) **nllb**: NLLB-200-distilled-600M [NLLB Team et al., 2022], the standard dedicated low-resource MT baseline; language-variety codes were assigned per item during validation, and every item received the Dari code (prs_Arab) — the pilot contains no Iranian Farsi source items, so Iranian Farsi appears below only as an unwanted output register. Unlike systems B–F, NLLB receives the utterance alone (it cannot condition on context) and decodes with beam 2 capped at 120 new tokens. (B) **frontier naive**: Kimi K3, the largest open instruct model on the Fireworks serverless catalogue at pin time (2026-08-10), prompted only to translate. (C) **frontier audience**: the same model with a system prompt stating who is writing to whom, their relationship, and the live situation. (D) **frontier instructional**: C plus explicit guidance on taarof mechanics, indirectness, and face, plus three few-shot exemplars from the held-out split. (E, F) **small naive** / **small audience**: gpt-oss-20b, a self-hostable 20B model, under the B and C prompts. All calls at temperature 0, cached, with measured token counts.

Failure class	Items	Judged	Pres.	Degr.	Inv.
indirect_refusal_as_assent	13	36	0.056	0.861	0.083
intent_inversion	10	42	0.167	0.762	0.071
obligation_face_deleted	10	30	0.267	0.667	0.067
register_violation	12	36	0.194	0.694	0.111
ritual_refusal_literal	15	36	0.361	0.417	0.222

Table 1: Failure-class composition (items of 60) and pooled human-judgment outcomes over the judged subset (Judged = judgments, 6 systems $\times$ judged items of that class; Pres./Degr./Inv. = shares of judgments rated 3/2/1). Aggregated from the per-system table in the benchmark release.

4.2 Evaluation protocol

Blind native-speaker judgment on a three-point intent scale (3 preserved, 2 degraded, 1 inverted or deleted) and a seven-point appropriateness scale, with optional notes and quick-flags (Iranian register, broken syntax, wrong language). The protocol follows the two-level design of [Tenzer et al. \[2026\]](#), pairing text-level analysis with recipient evaluation, and adapts the core-category coding of [House and Kádár \[2021\]](#) to ground the failure taxonomy. A stratified random subset of 30 items (15 per direction) was judged across all six systems, 180 judgments, system identity hidden and order randomized, by a single native Dari-speaking judge (the first author). The subset balances the two directions by design, so pooled rates weight the harder en2fa direction more heavily than its 20-of-60 share of the corpus; per-direction results are reported separately throughout. The two pre-specified questions, the equal-allocation direction split, and the 80% judge-calibration bar were fixed in the study plan before judging began (they are recorded in the repository history); we did not file an external preregistration; the randomized blind judging order is released with the benchmark. Two LLM judges were calibrated against these labels under a pre-specified 80% exact-agreement bar with one permitted rubric iteration: a scripted DeepSeek v4-pro judge and a panel of Claude Fable 5 agents. Both failed calibration (Section 5.5), so machine labels are reported only as an agreement finding and every evaluation number in this paper is human-labeled.

4.3 Statistical analysis

With 30 paired judgments per system, the study is powered only for large effects, and we report statistics accordingly [\[Card et al., 2020\]](#). For the two pre-specified comparisons we use exact tests: the exact (binomial) McNemar test on discordant pairs for the paired instructional-vs-audience comparison, since all systems are judged on the same 30 items [\[Dietterich, 1998\]](#), and Fisher’s exact test for the between-direction comparison, which is unpaired. Judgments are clustered within items (six systems per item), so for the direction comparison we also report an item-level test that collapses each item to a binary outcome. Headline preservation rates carry Wilson 95% score intervals, which remain valid at small n [\[Newcombe, 1998\]](#). Appropriateness is ordinal, so paired system comparisons on it use the exact sign test rather than tests on means [\[Howcroft and Rieser, 2021\]](#). All other pairwise contrasts among the six systems are exploratory and reported descriptively, without correction; the analysis script is released with the benchmark.

System	n	Preserved	Degraded	Inverted	Approp.
A nllb-600M	30	0.133	0.767	0.100	3.47
B kimi-k3 naive	30	0.200	0.667	0.133	3.40
C kimi-k3 audience	30	0.167	0.767	0.067	4.73
D kimi-k3 instructional	30	0.300	0.533	0.167	4.57
E gpt-oss-20b naive	30	0.200	0.667	0.133	4.67
F gpt-oss-20b audience	30	0.233	0.700	0.067	4.93

Table 2: Intent preservation and mean appropriateness by system; 180 blind human judgments over a stratified 30-item subset. Rates are shares of judgments rated 3 (preserved), 2 (degraded), and 1 (inverted or deleted).

5 Results

5.1 No system reliably preserves intent

Table 2 reports the headline. The best configuration, the frontier model with explicit cultural instruction, fully preserved intent in 30.0% of judgments. Every other configuration fell between 13.3% and 23.3%. The dominant outcome everywhere is degradation: the message arrives, the force does not. The dedicated MT baseline is worst on preservation (13.3%) and near-worst on appropriateness (3.47, ahead only of the naive frontier prompt at 3.40), consistent with the deployed default for these language pairs deleting pragmatic content as a matter of course. At $n = 30$ judgments per system the Wilson 95% intervals on preservation are wide and overlapping (D: [.17, .48]; F: [.12, .41]; A: [.05, .30]), so the system ranking should be read as provisional; what the intervals do establish is the headline ceiling, since even the upper bound of the best system falls below half of judgments preserved.

5.2 Cultural instruction adds both tails

Pre-specified question 1 asked whether explicit instruction (D) beats audience-targeting alone (C) on intent, given that Tenzer et al. [2026] found no significant recipient-side gain from instruction for appropriateness. Descriptively, instruction moves both tails at once: D nearly doubles full preservation over C (preservation 30.0% vs 16.7%) while also more than doubling catastrophic inversion (inversion 16.7% vs 6.7%), and C retains slightly higher appropriateness (4.73 vs 4.57). None of these paired differences reaches significance at this sample size (exact McNemar on preservation: 8 vs 4 discordant items, $p = .39$; on inversion: 5 vs 2, $p = .45$; exact sign test on appropriateness: $p = .66$), so the pilot cannot rule out a null effect of instruction; what it can rule out is the expectation that instruction only helps, since the inversion tail grows in the same configuration whose preservations grow. The two tails have different mechanisms, however: all five of D’s inverted-or-deleted judgments are en2fa generation failures (four flagged broken-syntax or wrong-language, one an incomplete translation), not pragmatic overshoots. Instruction’s gains are pragmatic, while its added catastrophic tail is generation breakdown under the heavier prompt, the failure class examined in Section 5.3. The threshold effect of Tenzer et al. [2026] thus survives on the appropriateness axis, while on the intent axis instruction helps where the model can generate fluent Dari and coincides with breakdown where it cannot. If the pattern holds at larger n , it argues for adaptive context layers, audience-targeting by default and heavy instruction only where generation is robust enough to carry it, rather than a single maximal prompt.

System	Dir.	n	Pres.	Degr.	Inv.	Appr.
A nllb	en2fa	15	0.133	0.800	0.067	4.00
A nllb	fa2en	15	0.133	0.733	0.133	2.93
B naive	en2fa	15	0.067	0.733	0.200	3.20
B naive	fa2en	15	0.333	0.600	0.067	3.60
C audience	en2fa	15	0.267	0.600	0.133	4.13
C audience	fa2en	15	0.067	0.933	0.000	5.33
D instruct.	en2fa	15	0.200	0.467	0.333	3.93
D instruct.	fa2en	15	0.400	0.600	0.000	5.20
E small naive	en2fa	15	0.200	0.600	0.200	4.73
E small naive	fa2en	15	0.200	0.733	0.067	4.60
F small aud.	en2fa	15	0.267	0.667	0.067	5.00
F small aud.	fa2en	15	0.200	0.733	0.067	4.87

Table 3: Intent outcomes and mean appropriateness by system and direction (Pres./Degr./Inv. = shares of judgments rated 3/2/1; Appr. = mean appropriateness; systems A–F as in Table 2). Table 4 gives the validator flags underlying the generation-breakdown reading of en2fa failures.

System	n	Iranian register	Broken syntax	Wrong language
A nllb-600M	30	7	2	0
B kimi-k3 naive	30	1	25	8
C kimi-k3 audience	30	0	7	3
D kimi-k3 instructional	30	0	6	4
E gpt-oss-20b naive	30	1	1	2
F gpt-oss-20b audience	30	4	0	0

Table 4: Validator quick-flags per system (counts over 30 judgments each, both directions pooled). The dedicated MT baseline drifts to Iranian register; the naive frontier prompt dominates broken-syntax and wrong-language flags.

5.3 The asymmetry reversed: generation, not comprehension, is the bottleneck

Pre-specified question 2 predicted that inversion would concentrate in the Dari-to-English (fa2en) direction, where high-context force must be recovered by inference. The data reversed it: inversion or deletion occurred in 16.7% of English-to-Dari (en2fa) judgments versus 5.6% of Dari-to-English (15/90 vs 5/90; Fisher’s exact test, $p = .031$; Table 3 and Figure 2). Because the six judgments on each item are not independent, we also test at the item level: 11 of 15 en2fa items drew a rating of 1 from at least one system, versus 3 of 15 fa2en items (Fisher’s exact test, $p = .009$), so the asymmetry is not an artifact of a few items failing under every system. Decomposition shows the reversal is driven by a failure class our pre-specified hypothesis did not separate: of the 15 en2fa judgments rated 1, ten carry broken-syntax or wrong-language flags and four more are empty or incomplete outputs by the validator’s notes, leaving a single purely lexical-pragmatic failure; the naive frontier configuration alone drew broken-syntax flags on 25 of its 30 judgments across both directions (Table 4). Meanwhile the fa2en direction almost never inverts but almost never fully lands either: under the frontier audience condition (C), 93.3% of fa2en judgments were rated degraded. The two directions fail in different classes. Into Dari, current models fail at the fluency layer before pragmatics is reached, consistent with Afghan Dari generation being undertrained rather than unactivated, the distinction Tenzer et al. [2026] could not probe for high-resource Japanese. Out of Dari, fluency survives and pragmatic force quietly dies, which is the failure mode of the motivating example.

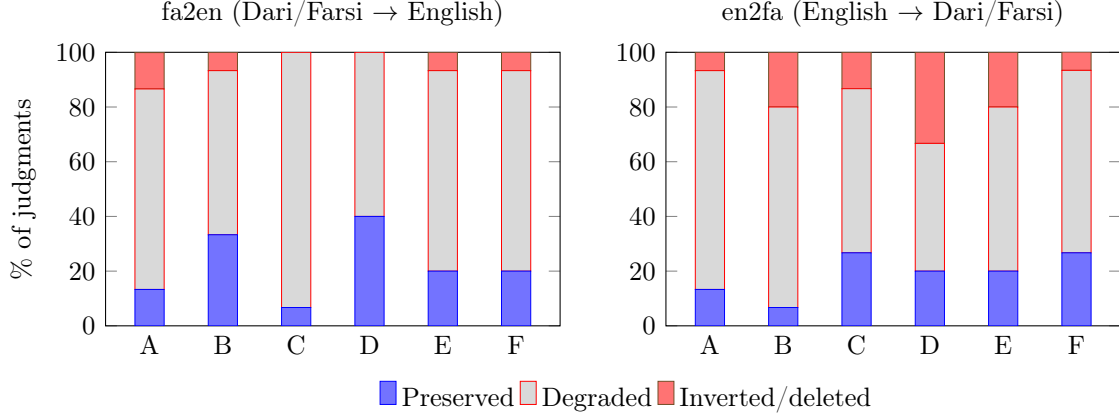


Figure 2: Intent outcomes by system and direction (shares of the 15 blind judgments per cell, recomputed from the released human judgments; same values as Table 3). Systems A–F as in Table 2. Out of Dari (left), inversion is rare and degradation dominates; into Dari (right), the inverted/deleted share grows in four of the five LLM configurations (B–E; unchanged for F), driven by generation breakdown rather than pragmatic inversion.

System	Basis	USD / 1k msgs
A nllb-600M	self-hosted est.	0.06
B kimi-k3 naive	API measured	10.32
C kimi-k3 audience	API measured	10.55
D kimi-k3 instructional	API measured	7.87
E gpt-oss-20b naive	API measured	0.10
F gpt-oss-20b audience	API measured	0.22
E/F gpt-oss-20b	self-hosted est.	0.42

Table 5: Cost per 1,000 successful messages. API rows use measured tokens at pinned prices; self-hosted rows assume \$2.50 per A100-class GPU hour with stated throughput assumptions. Failed calls (4 of 108 small-system calls produced no translation) are excluded from the cost basis and scored as deleted intent where judged.

5.4 A 20B model with a one-line wrapper is not separable from the frontier model at 2–3% of the cost

The small audience configuration (F) reached 23.3% preservation with tied-lowest inversion (6.7%) and the highest appropriateness in the study (4.93), at \$0.22 per 1,000 messages of measured API cost versus \$7.87 to \$10.55 across the three frontier configurations, and an estimated \$0.42 self-hosted (Table 5). Exploratory paired tests find no detectable difference between F and either frontier configuration on preservation (exact McNemar vs C: $p = .69$; vs D: $p = .79$) or appropriateness (exact sign test vs C: $p = .52$; vs D: $p = 1.0$); with 30 paired judgments these tests can only detect large gaps, so this is a failure to separate the systems, not evidence of equivalence. The honest cost-side claim is narrower and still consequential: the pilot provides no evidence that the like-for-like audience-prompt price difference of roughly $47\times$ (\$10.55 vs \$0.22 per 1,000 messages) buys any intent preservation, and the value observed so far concentrates in the context layer and the native-validated data that measures it. Total API spend for the whole pilot, translation and judging included, was \$3.83.

5.5 Native-speaker judgment is not yet automatable for this task

Against the pre-specified 80% exact-agreement bar, the scripted DeepSeek judge reached 39.4% exact intent agreement under its first rubric and 48.3% under the one permitted revision ($n = 178$ of 180: two outputs were unparseable, both on broken-output items the human rated 1; scoring them as disagreements gives 47.8%). The Claude Fable 5 panel reached 48.9% ($n = 180$) under a single rubric, so the two arms had unequal tuning budgets. Raw exact agreement flatters both judges: the human labels are 68.3% “degraded”, so a constant-label judge would score 68.3%, and chance-corrected agreement is $\kappa = 0.14$ (DeepSeek) and $\kappa = 0.17$ (Fable); within-one-point agreement (96.6% and 99.4%) and appropriateness correlation ($r = 0.38$ and $r = 0.63$) show the judges track coarse quality while missing the category boundaries. The disagreement is diffuse rather than a single leniency offset, though the largest single stream is degraded-read-as-preserved: judges credit gist transfer where the native judge docks for meaning-bearing lexical choices (Iranian *faktor* where Afghan Dari uses *bel* for invoice), dialect drift in load-bearing words, and softened force. Two caveats bound this finding: the Fable panel judged outputs against items whose reference renderings the same model family drafted, and it ran through an interactive session rather than a metered API, so that arm is not reproducible from the API key alone; and with a single annotator we cannot separate “the task is hard for judges” from “the rubric is underspecified” from “these labels are one speaker’s idiolect” — the second-annotator study is the discriminating experiment. We conclude only that two off-the-shelf judge configurations did not reproduce this annotator’s labels; all evaluation numbers in this paper are therefore human-labeled, and we release the machine judgments as calibration data.

6 Discussion

Three implications. First, for deployment: the pilot’s best cost-quality point is a small self-hostable model behind a thin audience-context layer, which reframes culturally competent translation as a data and product problem, per-culture context packs and native-validated benchmarks, rather than exclusively a frontier-model problem. On this pilot the cheapest gains came from the context layer; we did not test fine-tuning, which [Gohari Sadr et al. \[2025\]](#) report as substantially more effective for in-culture behaviour, so we make no claim about the relative return of the two routes. Second, for the science of cultural NLP: the direction decomposition suggests low-resource cultural competence has two separable bottlenecks, generation fluency (absent capability) and pragmatic transfer (unactivated capability), and benchmarks that do not separate them will misattribute one to the other. Third, for evaluation methodology: chance-corrected judge agreement of $\kappa \leq 0.17$ on a task where a native judge produced decisive labels suggests intent preservation belongs, for now, to the class of properties that must be measured by the affected community itself, an argument for participatory benchmark construction in exactly the languages least represented in training data.

7 Limitations

This is a pilot. The judged subset is 30 items and 180 judgments; the validator, judge, and first author are the same person, and no inter-annotator agreement can be reported yet; a second native Dari annotator on a subset is the immediate next step, and intra-annotator test-retest reliability is also unmeasured. Judgments are blind to system identity but not to item provenance: the judge authored or validated every item, including its reference renderings, so “blind” is narrower here than in a naive-recipient design such as [Tenzer et al. \[2026\]](#), whose independent recipient panel we do not

have. Thirty-two of sixty items are synthetic variants sharing pragmatic cores with their parents, and the synthetic share is direction-skewed: the en2fa arm, which carries the direction finding, contains no author-seed items (14 of its 20 items are synthetic). The six configurations also differ in more than model and prompt: NLLB receives no context and decodes under a length cap (one of its scored failures is a truncation), and the two LLM families ran under different reasoning-token budgets; model, input condition, and decoding budget are confounded. Runs were single-pass at temperature 0, so run-to-run variance on serverless serving is unmeasured. An internal adversarial audit of the blind judging batches noted stylistic tells (headers, leaked deliberation) that could in principle hint at the model family, though never the system label. Kimi K3 sometimes emits visible deliberation in its output field, judged verbatim as delivered text. Four small-model calls produced no output at any reasoning-token budget; three fell in the judged subset and are scored as deleted intent. Beyond the two pre-specified comparisons, all pairwise system contrasts are exploratory and uncorrected for multiple comparisons, and judgment-level tests inherit the item clustering described in Section 4.3. The Chinese arm of the intended triangle is designed but unbuilt, and all claims are scoped to trade-register Dari/Farsi and English.

8 Ethical Considerations

Dari and Hazaragi materials involve communities under active persecution. All benchmark items are fictional and contain no real names, businesses, or identifying details; the motivating example is de-identified and used with the consent of the conversation’s owner [TBD: confirm written consent before release]. The single-annotator design concentrates interpretive authority in one speaker of one dialect; we scope claims accordingly and invite community annotation in the public release.

9 Conclusion

SAWDA turns a familiar frustration, the translated message that says the words and betrays the meaning, into a measured phenomenon: on native-validated trade communication, no system we tested preserved speaker intent in more than 30% of judgments (best 9 of 30, 95% CI 17–48%); cultural instruction’s gains arrived together with generation breakdown; the failures for Dari concentrated in generation rather than comprehension; and neither of two LLM-judge families reproduced the native judge’s labels. The benchmark, code, translations, and all 180 human judgments are released at [TBD: repo and dataset links] to make the next claim about culturally competent translation a checkable one.

Acknowledgments

[TBD: Jonathan Leigh for the motivating problem, the real-world example, and Chinese-side verification in the planned extension; R. Ramanujan for advising; confirm roles and consent before release.]

References

- William O. Beeman. *Language, Status, and Power in Iran*. Indiana University Press, 1986.
- Penelope Brown and Stephen C. Levinson. *Politeness: Some Universals in Language Usage*. Cambridge University Press, 1987.

- Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998.
- Nikta Gohari Sadr, Sahar Heidariasl, Karine Megerdooimian, Laleh Seyyed-Kalantari, and Ali Emami. We politely insist: Your LLM must learn the Persian art of taarof. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2509.01035.
- H. Paul Grice. Logic and conversation. In *Syntax and Semantics 3: Speech Acts*. Academic Press, 1975.
- Edward T. Hall. *Beyond Culture*. Anchor Books, 1976.
- Geert Hofstede. *Culture’s Consequences: Comparing Values, Behaviors, Institutions and Organizations Across Nations*. Sage Publications, 2001.
- Wei Hong and Chao-Mei Tu. Business culture, etiquette, and intercultural communication: Exploring theories of pragmatics in business Chinese pedagogy. In Li-Yu Chen and Chao-Mei Tu, editors, *New Developments in Chinese for Specific and Professional Purposes*. Springer, Singapore, 2026. doi: 10.1007/978-981-95-5315-0_1.
- Juliane House and Dániel Z. Kádár. *Cross-Cultural Pragmatics*. Cambridge University Press, 2021.
- David M. Howcroft and Verena Rieser. What happens if you treat ordinal ratings as interval data? Human evaluations in NLP are even more under-powered than you think. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- Kwang-kuo Hwang. Face and favor: The Chinese power game. *American Journal of Sociology*, 92(4), 1987.
- Youngeun Koo, Jiwoo Lee, Dojun Park, Seohyun Park, and Sungeun Lee. Evaluating large language models on understanding Korean indirect speech acts. *arXiv preprint arXiv:2502.10995*, 2025.
- Sofia A. Koutlaki. Offers and expressions of thanks as face enhancing acts: Tæ’arof in Persian. *Journal of Pragmatics*, 34(12):1733–1756, 2002.
- Erfan Moosavi Monazzah, Vahid Rahimzadeh, Yadollah Yaghoobzadeh, Azadeh Shakery, and Mohammad Taher Pilehvar. PerCul: A story-driven cultural evaluation of LLMs in Persian. *arXiv preprint arXiv:2502.07459*, 2025. Published at NAACL 2025.
- Junho Myung, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty, Eunsu Kim, Carla Perez-Almendros, Abinew Ali Ayele, et al. BLENd: A benchmark for LLMs on everyday knowledge in diverse cultures and languages. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024. arXiv:2406.09948.
- Robert G. Newcombe. Two-sided confidence intervals for the single proportion: Comparison of seven methods. *Statistics in Medicine*, 17(8):857–872, 1998.
- NLLB Team, Marta R. Costa-jussà, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.

- Zahra Pourbahman, Fatemeh Rajabi, Mohammadhossein Sadeghi, Omid Ghahroodi, Somaye Bakhshaei, Arash Amini, Reza Kazemi, and Mahdiah Soleymani Baghshah. ELAB: Extensive LLM alignment benchmark in Persian language. *arXiv preprint arXiv:2504.12553*, 2025.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, 2023.
- Abhinav Rao, Akhila Yerukola, Vishwa Shah, Katharina Reinecke, and Maarten Sap. NormAd: A framework for measuring the cultural adaptability of large language models. *arXiv preprint arXiv:2404.12464*, 2024. Published at NAACL 2025.
- Laura Ruis, Akbir Khan, Stella Biderman, Sara Hooker, Tim Rocktäschel, and Edward Grefenstette. Large language models are not zero-shot communicators. *arXiv preprint arXiv:2210.14986*, 2022.
- Jiaxing Sun, Weiquan Huang, Jiang Wu, Chenya Gu, Wei Li, Songyang Zhang, Hang Yan, and Conghui He. Benchmarking Chinese commonsense reasoning of LLMs: From Chinese-specifics to reasoning-memorization correlations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11205–11228, 2024. arXiv:2403.14112.
- Helene Tenzer, Oumnia Abidi, and Stefan Feuerriegel. Prompting for pragmatics: Improving the cultural sensitivity of LLM translations for business emails. *Management International Review*, pages 1–37, 2026. doi: 10.1007/s11575-026-00637-4.
- Chen Zhang, Mingxu Tao, Zhiyuan Liao, and Yansong Feng. MiLiC-Eval: Benchmarking multilingual LLMs for China’s minority languages. *arXiv preprint arXiv:2503.01150*, 2025. Findings of ACL 2025.